

Datasets for Modi → Marathi Transliteration: A Deep Dive

Project: मोडी ते मराठी (modi-to-Marathi) **Repository:** <https://github.com/lgtkgtv/ai4-transliteration-modi-to-marathi> **Author:** Sachin · **Audience:** the data/engineering team **Purpose:** Catalogue every dataset relevant to training or fine-tuning a Modi→Marathi transliteration model, with honest notes on what each one actually contains, how to get it, and its licence.

0. The one distinction that organizes everything

Before the list, internalize this — it prevents wasted effort:

Recognition data ≠ transliteration data. Most Modi datasets are **character recognition** sets (isolated letters with a class label). Our task needs **paired transliteration** data (an image of Modi text → the correct Marathi/Devanagari text). Recognition sets are still useful — but mainly for pretraining and synthetic data, not for directly learning end-to-end transliteration.

So we sort everything into three buckets:

Bucket	What it gives us	Example
A. Paired transliteration	Directly trains/evaluates our exact task	MoDeTrans
B. Modi recognition / raw images	Pretraining a vision encoder; raw pages to label; synthetic word building	MODI-HChar, MODI-HHDoc
C. Adjacent Indic transliteration text	The craft of transliteration modeling; clean Marathi target text	Aksharantar, Dakshina

1. Bucket A — the dataset that matches our exact task

MoDeTrans ☆ (start here)

- **What:** ~2,043 real handwritten Modi **document images**, each paired with an **expert-verified Devanagari transliteration**. Spans Shivakalin, Peshwekalin, Anglakalin. Split 80:10:10.
 - **Why it's special:** the **only** public dataset that gives us image → Devanagari text ground truth at the document level — exactly our task.
 - **Access:** Hugging Face, `historyHulk/MoDeTrans` (public). Load with the `datasets` library.
 - **Companion:** `historyHulk/SynthMoDe` — synthetic Modi images rendered from the same Devanagari text (using Modi fonts). Public. Great for augmentation and for a "train-on-real / test-on-synthetic" generalization check.
 - **Caveat:** ~2k examples is small for deep learning. Expect to augment it heavily (Bucket B + C strategies below).
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2. Bucket B — Modi recognition & raw-image datasets

These come mostly from the prolific **Deshmukh & Kolhe** group (Dr. B.A.M. University, Aurangabad) and **Jadhav & Inamdar**. They are excellent, but note the **research-only, no-redistribution** licences — we can train on them, but we generally **cannot republish** them or a thin derivative.

Dataset	Contents	Size	Access / ID	Licence
MODI-HChar	Isolated characters: 57 classes (10 numerals, 12 vowels, 35 consonants), 170×170 px grayscale JPG	~575,920 images	IEEE DataPort DOI 10.21227/v2kt-rr94 ; Mendeley <code>pk2zrt58pp</code>	Research-only, no redistribution
MODI-HHDoc	Whole historical Modi document page images	3,350 images	IEEE DataPort DOI 10.21227/1z10-	Research-only, no redistribution

Dataset	Contents	Size	Access / ID	Licence
	(Peshwa-era), 24 folders	(~6.83 GB)	w986 ; Mendeley sg337vf6wn	
Handwritten MODI Characters	Isolated character samples	—	IEEE DataPort DOI 10.21227/z3gg-8b29 (Jadhav & Inamdar)	IEEE DataPort subscription
Handwritten Modi Lipi Barakhadi	Barakhadi — consonant×vowel-sign combinations	—	IEEE DataPort	IEEE DataPort

Practical notes on access: - **Mendeley Data** mirrors of MODI-HChar and MODI-HHDoc are usually **downloadable without a subscription** (still under the research-only licence) — often the easiest route. IEEE DataPort frequently requires an IEEE/DataPort subscription. - **MODI-HHDoc gives us raw pages but not transliterations.** It's a superb source of real document images onto which our experts can add Devanagari labels — effectively a way to grow Bucket A. - **Character sets (MODI-HChar etc.) are recognition data.** Their best use for us is (a) **pretraining the vision encoder** so it already "knows" Modi letter shapes, and (b) **assembling synthetic words/lines** by stitching characters.

3. Bucket C — adjacent Indic transliteration corpora (not Modi, still valuable)

These are **Roman ↔ native-script** text datasets. They contain **no Modi**, so they can't teach Modi letter shapes. But they're valuable in two concrete ways: they teach the **modeling craft** of transliteration, and they are a large, clean source of **Marathi Devanagari text** — which we can feed into Modi fonts to mass-produce **synthetic Modi→Devanagari pairs**.

Dataset	Contents	Marathi?	Size	Access	Licence
		Yes			CC-BY (manual) /

Dataset	Contents	Marathi?	Size	Access	Licence
Aksharantar (AI4Bharat)	Roman↔Indic word pairs, 21 languages		~ 26M pairs	HF ai4bharat/ Aksharantar	CC0 (mined)
Dakshina (Google Research)	Native-script Wikipedia text + romanization lexicon + parallel sentences , 12 South Asian languages	Yes (mr)	~ 300K word pairs, ~ 120K sentence pairs	GitHub google- research- datasets/ dakshina	CC BY-SA 4.0

Why we care, concretely: - **Dakshina's Marathi native-script Wikipedia text and sentence pairs** = a ready supply of correct modern Marathi (Devanagari) sentences. Render those through **MarathiCursive / Noto Sans Modi** → instant synthetic Modi→Devanagari training pairs. - **Aksharantar** is the proof that data scale + a small model wins at transliteration; its Marathi portion and methodology are a useful reference even though the direction (Roman↔Devanagari) differs from ours (Modi→Devanagari).

4. Tools & libraries worth adopting

Tool	Use to us
Hugging Face datasets / transformers	Load MoDeTrans/SynthMoDe; run/fine-tune models
Modi fonts: MarathiCursive, Noto Sans Modi	Generate synthetic Modi images from Devanagari text
Indic NLP Library	Devanagari normalization, script conversion, tokenization for the target side

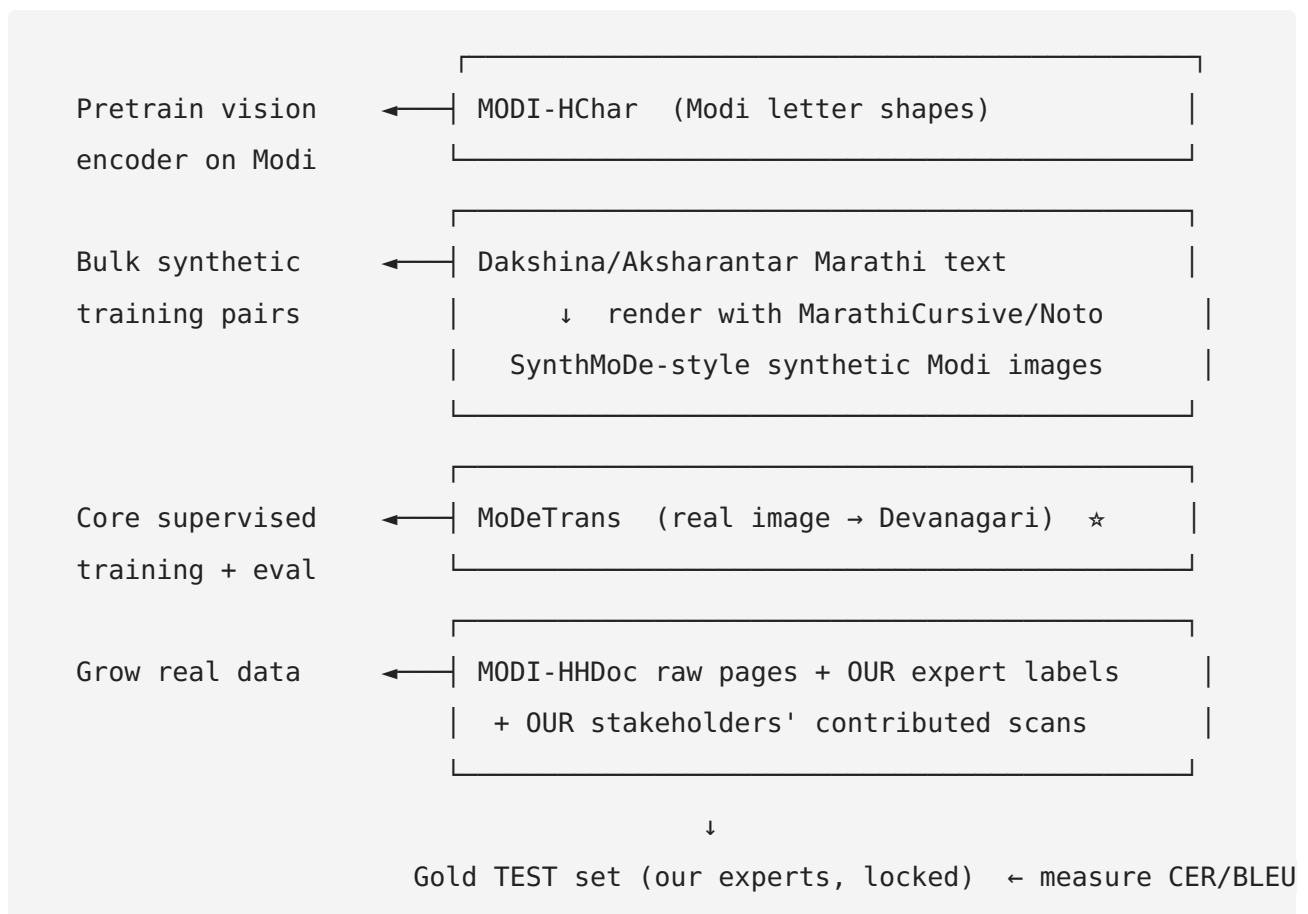
Tool	Use to us
indic-transliteration (Sanscript)	Convert Devanagari ↔ romanization schemes (useful for evaluation/normalization)
IndicXlit (AI4Bharat)	Reference model + library for Indic transliteration; baseline ideas
OpenCV	Preprocessing real document scans (denoise, threshold, deskew)

5. Licensing & ethics summary (read before training)

Source	Can we train on it?	Can we redistribute it?	Note
MoDeTrans / SynthMoDe	Yes	Check the HF card terms	Public; cite the paper
MODI-HChar / MODI-HHDoc	Yes (research)	No	Research-only; credit Deshmukh & Kolhe
Jadhav & Inamdar (IEEE DataPort)	Yes (per subscription)	No	Subscription/licence bound
Aksharantar	Yes	Yes (per CC-BY/CC0)	Attribute appropriately
Dakshina	Yes	Yes (CC BY-SA 4.0)	Share-Alike — derivatives must use the same licence

Two stakeholder takeaways: 1. If we ever **publish our own dataset**, we must build it from sources whose licences permit redistribution (our own scans + expert labels, synthetic data, and CC-licensed text) — **not** from the no-redistribution Modi sets. 2. Always record **provenance** per item so licence and credit obligations are traceable.

6. How each dataset maps to our pipeline (the synthesis)



The strategy in one line: **pretrain on character shapes, bulk up with synthetic pairs from licence-clean Marathi text, train and measure on MoDeTrans, and grow real data from MODI-HHDoc pages plus our own contributed scans — all verified by our experts.**

7. Recommended starting bundle (week 1–2)

1. **Pull** `historyHulk/MoDeTrans` + `historyHulk/SynthMoDe` from Hugging Face — this is our backbone and baseline.
 2. **Download MODI-HChar and MODI-HHDoc from the Mendeley mirrors** (research-only) — for vision-encoder pretraining and as raw pages to label.
 3. **Grab Dakshina (mr)** from GitHub — a licence-clean source of Marathi Devanagari text for synthetic generation.
 4. **Install** the Modi fonts + Indic NLP Library + OpenCV.
 5. **Build the gold test set first** (see `docs/03_Dataset-Preparation-Plan.md`) before training anything.
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8. Sources & access links

- **MoDeTrans / SynthMoDe** — Hugging Face, org `historyHulk`. Described in arXiv:2503.13060 / ICDAR 2025 (DOI 10.1007/978-3-032-04630-7_3).
- **MODI-HChar** — Deshmukh, M. & Kolhe, S. (2023). IEEE DataPort DOI **10.21227/v2kt-rr94**; Mendeley Data `pk2zrt58pp` (<https://data.mendeley.com/datasets/pk2zrt58pp/1>).
- **MODI-HHDoc** — Deshmukh, M. & Kolhe, S. (2023). IEEE DataPort DOI **10.21227/1z10-w986**; Mendeley Data `sg337vf6wn` (<https://data.mendeley.com/datasets/sg337vf6wn/1>).
- **Handwritten MODI Characters** — Jadhav, S. & Inamdar, V. (2021). IEEE DataPort DOI **10.21227/z3gg-8b29**.
- **Handwritten Modi Lipi Barakhadi** — IEEE DataPort (<https://ieee-dataport.org/documents/handwritten-modi-lipi-barakhadi-dataset>).
- **Aksharantar** — AI4Bharat. Hugging Face `ai4bharat/Aksharantar`; paper "Aksharantar: Towards building open transliteration tools for the next billion users." Licences CC-BY / CC0.
- **Dakshina** — Google Research. GitHub `google-research-datasets/dakshina`; Roark et al., LREC 2020. Licence CC BY-SA 4.0.
- **IndicXlit** — AI4Bharat. GitHub `AI4Bharat/IndicXlit`; `pip install ai4bharat-transliteration`.
- **Underlying Modi character-recognition lineage** — Deshmukh, Patil & Kolhe (2015), "Off-line Handwritten Modi Numerals Recognition using Chain Code"; Chandure & Inamdar (2021), "Offline handwritten MODI character recognition using GoogLeNet and AlexNet," ICCMS.

Verify dataset sizes, DOIs, and licence terms on the source pages before formal use — repositories update over time.